

AegisPay — Build Errors & How We Solved Them

Git Error aur uske Solution. Har naye error neeche add karo (points 4 ko chhuro)

PDF dobara banane ke liye: `python pdf/build_error_guide_pdf.py`

1. Docker app: ModuleNotFoundError: No module named 'api'

- Error: app container crash loop — No module named 'api'.
- Problem: Build context api/ tha, lekin package repo root par hai; COPY . . se modules /app mein flat ho gaye aur import api fail.
- Fix: Context = repo root, aur COPY api ./api kar phir pip install.

2. Migration: connection to server at "db" ... Connection refused

- Error: db_migrate started before Postgres ready; connection refused, exit 2.
- Problem: db par koi healthcheck nahi tha; race condition.
- Fix: db healthcheck add karo + depends_on: db: condition: service_healthy.

3. Migration: fe_sendauth: no password supplied

- Error: psql se no password supplied.
- Problem: TCP par password required tha, container mein PGPASSWORD nahi tha.
- Fix: PGPASSWORD: aegispay env mein set karo (+ ON_ERROR_STOP=1).

4. Localstack: exit code 55 ... License activation failed

- Error: LocalStack quit, "No credentials / license".
- Problem: localstack: latest ab auth/license maangta hai.
- Fix: Free OSS image pin karo — localstack/localstack:3.5.0.

5. Docker build: pip ... ReadTimeoutError ... pythonhosted

- Error: Image build mein pip download timeout.
- Problem: Default pip timeout chota tha, network slow.
- Fix: Dockerfile mein PIP_DEFAULT_TIMEOUT=120 + --timeout 120.

6. CI test: '_IncludedRouter' object has no attribute 'path'

- Error: Smoke test fail — 1 failed, 40 passed.
- Problem: Test app.routes par route.path padh raha tha; FastAPI 0.141.1 (latest) internal _IncludedRouter deta hai. Local had 0.135.2 -> local pass, CI fail.
- Fix: app.openapi()["paths"] use karo (stable). FastAPI pin: >=0.115, <0.142.

7. PDF renderer: saw instead of expected </i>

- Error: ReportLab parse error while rendering inline code/emphasis.
- Fix: Code spans ko placeholder se replace karo (pehle), phir bold/italic, phir restore.

8. PDF build: ModuleNotFoundError: No module named 'pdf'

- Error: python pdf/build_x.py chalane par pdf module nahi mila.
- Fix: Script ke top par repo root sys.path mein add karo:
`sys.path.insert(0, str(Path(__file__).resolve().parents[1]))`.

9. CI gitleaks: failed to scan Git repository — "stderr is not empty"

- Error: CI test job fail — gitleaks secret scan abort.

- Problem: gitleaks HEAD ko uske parent commit se diff karta hai, par PR default shallow checkout (depth 1) se aata hai jisme parent commit nahi hota -> ambiguous argument '<sha>^<sha>': unknown revision.
- Fix: checkout mein fetch-depth: 0 (full history) + gitleaks step ko continue-on-error: true (report-only, koi real leak nahi mila).

10. CI integration: password authentication failed for user "aegispay_app"

- Error: Integration tests fail — InvalidPasswordError.
- Problem: CI ka bare Postgres service db/init.sql nahi chalata (wo sirf local compose entripoint mein hota hai), isliye aegispay_app role banti hi nahi -> migration ke grants + tests connect fail.
- Fix: Integration workflow mein migration se pehle psql -f api/db/init.sql chalao (role create + grants).

11. Auth: valid member token par /v1/me -> 403

- Error: E2E test — authenticated member ko /v1/me 403 insufficient role.
- Problem: Endpoint has_permission(principal.role, "member") check kar raha tha, lekin "member" ek role hai, permission nahi. Member role ke paas catalog.read hai, member nahi -> hamesha deny. Koi bhi authenticated user apni identity dekh sakta hai, is check ki zaroorat hi nahi.
- Fix: /v1/me se galt permission gate hataya (auth hi kaafi hai); RBAC gating domain routers hota hai.

12. Auth: bad JWT signature -> 500 (should be 401)

- Error: E2E test — invalid/expired token par /v1/me 500 INTERNAL_ERROR.
- Problem: resolve_principal mein verify_jwt ValueError raise karta hai (bad signature/expired), par wo catch nahi tha -> generic 500 handler tak pahunch jaata tha.
- Fix: resolve_principal mein ValueError catch karke HTTPException(401, "invalid token") raise karo — bad/expired token ab 401 deta hai.

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13. Core API: inconsistent error envelope (401 me {"detail":...})

- Error: E2E — auth error GET /v1/me no-token -> {"detail": "authentication required"} (FastAPI default), baaki errors {"code", "message", "request_id", "retryable"}. API ka error contract consistent nahi tha.
- Problem: resolve_principal rahi HTTPException(401); handlers sirf AppError ko envelope dete the, HTTPException/RequestValidationError ko nahi. Isliye HTTPException-based errors FastAPI ke default {"detail":...} format me aate the.
- Fix: exceptions.py me handlers add kiya — HTTPException (-> status-based code: 401->AUTHENTICATION_ERROR, 404->NOT_FOUND, ...) aur RequestValidationError (->VALIDATION_ERROR, 422). Ab sab errors same envelope dete hain. _http_code() helper se status->code mapping.
- Note (framework limitation): FastAPI 0.141 me router-level 404 (unknown route) aur uncaught 500 app handlers ke through nahi aate — wo FastAPI ke default response dete hain. Isliye consistency unit-test (_http_code) se prove ki gyi; 401/application errors E2E se.

14. Core API: /v1/readyz hamesha ready (DB check nahi) + /v1/live missing

- Error: /v1/readyz DB down hone par bhi {"status": "ready"} deta tha (hardcoded); /v1/live route hi nahi tha -> 404.
- Problem: Readiness probe sirf hardcoded string return karta tha; koi DB connectivity check nahi.
- Fix: /v1/readyz ab DB ko ping karta hai (select 1); DB nahi pahuncha toh 503 {"code": "NOT_READY"}. Aur naya GET /v1/live (200) add kiya.

15. Commerce build: NoReferencedTableError: Foreign key 'carts.agent_id' could not find table 'agents'

- Error: Phase 4 test run — SQLAlchemy threw NoReferencedTableError for carts.agent_id / orders.agent_id.
- Problem: ORM models mein agents table ke liye koi model tha hi nahi (migration mein table hai, par model nahi). SQLAlchemy FK ko resolve karne ke liye referenced table ka model metadata mein hona zaroori hai.
- Fix: models.py mein minimal Agent model add kiya (agents table map). (users FK ko model mein deliberately nahi rakha taaki woh bhi same error na de.)

16. Live API: product create par 500 INTERNAL_ERROR (nonexistent tenant)

- Error: Live E2E me /v1/products create -> 500 INTERNAL_ERROR.
- Problem: Test mein ek fabricated tenant_id use kiya jo tenants table mein exist nahi karta. Product insert me tenant_id -> tenants(id) FK violation ho gayi, jo unhandled IntegrityError -> generic 500.
- Fix: Ye code bug nahi tha (happy path integration-test me pass hai). Sahi tenant se live E2E 201/200 clean chal raha hai. Bas FH: real tenant ho. (Note: FK violation ab 500 deta hai — ideally 4xx hona chahiye, par ye edge case hai; generic 500 handler safely handles karta hai.)

17. Phase 5 build: column authorizations.updated_at does not exist

- Error: Authorization INSERT -> UndefinedColumnError: authorizations.updated_at does not exist.
- Problem: Authorization model TimestampMixin use kar raha tha, jo updated_at add karta hai. Par migration ke authorizations table mein sirf created_at hai (updated_at nahi). Model ↔ table mismatch.
- Fix: Authorization se TimestampMixin hata ke created_at explicitly map kiya (server_default). Ab INSERT sirf existing columns use karta hai.

18. Phase 5 build: approvals me NOT NULL expires_at missing

- Error: Approval INSERT -> NotNullViolationError: null value in column "expires_at" of relation "approvals".
- Problem: Approval model mein expires_at column map hi nahi tha, par table me not null hai.
- Fix: Model me expires_at (DateTime, not null) + created_at add kiya, aur service mein approve() par expires_at set karta hai.

19. Phase 5 test: approvals.approver_id FK violation (Key not present in users)

- Error: High-risk approve -> ForeignKeyViolationError: approvals_approver_id_fkey ... Key is not present in table "users".
- Problem: Approve test me random uuid4() approver_id use kiya, par DB me approvals.approver_id -> users(id) real FK hai — approver ko users me exist karna chahiye.
- Fix: Test helper _approvers(n) add kiya jo pehle users rows create karta hai (password_hash NOT NULL), phir un id se approve karta hai.

20. Phase 6: MemOutbox.emit() missing 1 required positional argument: 'payload'

- Error: Payment flow -> TypeError: MemOutbox.emit() missing 1 required positional argument: 'payload'.
- Problem: initiate_payment() ke default par outbox=MemOutbox tha — class pass ho rahi thi, instance nahi. Toh MemOutbox.emit(...) call par self drop ho jaata hai -> sirf 2 args milte.
- Fix: Default ko None karke andar outbox = outbox or MemOutbox() (instance) banaya.

21. Phase 6: payments INSERT me tenant_id NULL -> RLS new row violates row-level security

- Error: Payment save -> ProgrammingError: new row violates row-level security policy for table "payments" (tenant_id thodi null).
- Problem: DbPaymentAdapter.save() ORM Payment me tenant_id set hi nahi karta tha (TenantMixin required hai, par koi default nahi) -> NULL -> RLS WITH CHECK fail.

- Fix: DbPaymentAdapter ko tenant_id pass karke save() me tenant_id=uuid.UUID(...) set kiya.

22. Phase 6: payment re-fetch se payment not persisted

- Error: Router me payment create -> 500 "payment not persisted".
- Problem: Flow ka payment.id = uuid4().hex (32-char string), but DbPaymentAdapter.save() ORM Payment pe id set nahi karta tha, so DB row ko alag uuid4() id milti thi. Router outcome.payment_id se row dhoondta tha, milta nahi.
- Fix: save() me id=uuid.UUID(p.id) set kiya — ab DB row ki id == flow ki payment_id.

23. Phase 7 test: payments_order_id_fkey violation (payment insert)

- Error: Payment seed -> ForeignKeyViolationError: payments.order_id Key is not present in table "orders".
- Problem: Test me payment banate samay ek random order_id use kiya jo orders table me exists nahi karta; payment.order_id FK hai.
- Fix: Test helper ne tenant -> agent -> cart -> order -> payment ka pura chain seed kiya, phir payment ko real order_id se banaya.

24. Phase 7: reconciliation UNKNOWN me atka (commit nahi hua)

- Error: Reconcile ke baad payment 'UNKNOWN' hi raha (PAID nahi hua).
- Problem: reconcile_unknown() ne session.flush() kiya tha (change in-memory), lekin commit nahi kiya. Tab session close hone par change lost ho gaya -> next read par purana UNKNOWN mila.
- Fix: Service me await session.flush() ko await session.commit() se replace kiya (self-contained atomic op). Router path (DbSession dependency) waise bhi commit karta hai, double-commit harmless hai.

25. Phase 8: audit chain verify hamesha False (hash mismatch)

- Error: Append 2 events ke baad verify_chain() -> False, chain ko intact hona chahiye tha.
- Problem: Event ka event_hash ledger ke Python timestamp se compute hota hai, par DB created_at ko apne alag now() se set karta hai. Dono timestamps microsecond level par differ -> recompute/hash mismatch.
- Fix: append_audit me row.created_at = datetime.fromisoformat(event.created_at) set kiya — DB wahi timestamp store karta hai jo hash me use hua. Verify ab match karta hai.

26. Phase 8: audit_events INSERT se new row violates row-level security

- Error: App role INSERT INTO audit_events -> InsufficientPrivilegeError (RLS deny).
- Problem: audit_events RLS enabled hai par sirf SELECT policy tha. RLS enabled table me kisi command ke liye policy na ho toh wo deny all hota hai, isliye INSERT block ho jaata tha (grant hone ke baad bhi).
- Fix: Migration + live DB me create policy audit_insert on audit_events for insert with check (tenant_id = current_tenant()) add kiya — ab app role apne tenant ke andar append kar sakta hai (update/delete abhi bhi revoked).

27. Lint: B008 Do not perform function call Depends in argument defaults (FDP)

- Error: Depends(get_client) default arg par -> ruff B008.
- Problem: FastAPI purana style (default arg me Depends()) hai; ruff isko flag karta hai. Codebase already Annotated pattern use karta hai (DbSession = Annotated[..., Depends(get_session)]).
- Fix: Client = Annotated[ControlPlaneClient, Depends(get_client)] banake endpoint me client: Client use kiya. (Depends() default arg me nahi.)

28. Lint: ASYNC210 Async functions should not call blocking HTTP methods

- Error: await httpx.get(...) (blocking top-level httpx) -> ruff ASYNC210.
- Problem: httpx.get/post sync hai; async function me use nahi hota.
- Fix: httpx.AsyncClient use kiya (RazorpayAdapter jaisa) — await self._client.get/post.

29. Phase 10: campaign create 400 "margin below floor" (margin default 0)

- Error: POST /v1/campaigns -> VALIDATION_ERROR: margin below floor.
- Problem: check_caps me min_margin_pct=15 hai, aur default margin_pct=0 (caller ne set nahi kiya) -> 0 < 15 -> reject.
- Fix: Campaign create test me valid caps pass kiya (margin_pct: 20, duration_days: 7).

30. Phase 10: generate_opportunities me uuid.UUID(principal.subject) -> "badly formed hexadecimal UUID string"

- Error: POST /v1/opportunities/generate -> 500 ValueError: badly formed hexadecimal UUID string.
- Problem: agent_id=uuid.UUID(principal.subject) — token ka sub ("agent-1") ek valid UUID nahi hai.
- Fix: agent_id ko route me request body se lete hain (OpportunityGenerateIn.agent_id), subject se derive nahi karte.

31. Phase 11: buyer tools add_item/checkout allowlist me nahi the

- Error: Buyer flow me add_item/checkout use karna tha, par tools.registry.ALLOWED_TOOLS me sirf 3 the (discover/create_cart/request_authorization) -> buyer ke tools allow nahi honge.
- Problem: SELL buyer ko cart/order setup karna hai (add item, checkout). Ye money moves nahi hain (sirf cart/order setup), par allowlist me add nahi the — guardrail test fail hota.
- Fix: ALLOWED_TOOLS me add_item + checkout add kiya (cart/order setup). capture/refund/execute_payment ab bhi FORBIDDEN.

32. Phase 12: gateway router me Gateway.enter auth double-verify (reuse pattern)

- Error: Gateway route me token ko pehle verify_jwt karke claims (merchant_id/agent_id) nikalna padta hai, par Gateway.enter andar phir se authenticate(token) karta hai — double verify, par zaroori (claims chahiye before enter).
- Problem: Gateway.enter signature me authenticate + merchant_id/agent_id alag chahiye; router ko claims pehle chahiye.
- Fix: Router pehle verify(token) se claims nikalta hai, phir wahi token_gateway.enter ko pass karta hai (jo apne authenticate se subject jaght jaata hai). Redundant verify acceptable — Gateway unchanged reuse kiya.

33. Phase 14: reconciliation worker me Tenant model missing

- Error: Worker from api.db.models import Tenant -> ImportError: cannot import name 'Tenant'.
- Problem: ORM me tenants table ka koi model nahi tha (worker ko sari tenants enumerate karni hain; tenants table RLS nahi hota, isliye wo cross-tenant enumeration ka root hai).
- Fix: models.py me Tenant model add kiya (id, slug, name, currency, status). TimestampMixin ke baad define kiya (pehle NameError tha).

34. Phase 15 (red-team): FAKta ke liye webhook verification default weak (no secret)

- Error/Note: Bina RAZORPAY_WEBHOOK_SECRET configured, webhook router get_provider() (mock) par fallback karta hai jo har signature accept karta hai.
- Problem: Webhooks untrusted hain; prod me secret required hai. Dev/test me mock fallback ko treat karna theek hai, par prod me ye security gap hai.
- Fix: Red-team suite me isko document kiya. Production must set RAZORPAY_WEBHOOK_SECRET (router phir HMAC verify karta hai). No test gap — red-team suite attacks safely blocked.

Naya error yahan add karo (template)

```
## N. <short error title>
- **Error:** <paste error>
- **Problem:** <why it happened, 1-2 lines>
- **Fix:** <what we changed>
```